```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score
```

```
df=pd.read_csv('Medical Cost Dataset.csv')
```

```
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: Generate code with df New interactive sheet

```
df.tail()
```

	age	sex	bmi	children	smoker	region	charges
1333	50	male	30.97	3	no	northwest	10600.5483
1334	18	female	31.92	0	no	northeast	2205.9808
1335	18	female	36.85	0	no	southeast	1629.8335
1336	21	female	25.80	0	no	southwest	2007.9450
1337	61	female	29.07	0	yes	northwest	29141.3603

```
df.shape
```

```
(1338, 7)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       1338 non-null   int64  
 1   sex       1338 non-null   object 
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64  
 4   smoker    1338 non-null   object 
 5   region    1338 non-null   object 
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
df.isnull().sum()
```

	0
age	0
sex	0
bmi	0
children	0
smoker	0
region	0
charges	0

dtype: int64

```
df
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

1338 rows × 7 columns

Next steps: Generate code with df New interactive sheet

```
df=pd.read_csv('Medical Cost Dataset.csv')
x=df.drop('charges',axis=1)
y=df['charges']
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder

# Identify categorical and numerical columns
categorical_features = ['sex', 'smoker', 'region']
numerical_features = ['age', 'bmi', 'children']

# Create a column transformer for preprocessing
# Apply OneHotEncoder to categorical features and 'passthrough' numerical features
preprocessor = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features),
        ('num', 'passthrough', numerical_features)
    ],
    remainder='passthrough' # In case there are other columns, pass them through
)

# Fit and transform x_train
x_train_processed = preprocessor.fit_transform(x_train)

# Transform x_test (using the fitted preprocessor) for consistency, although not directly used in the fi
x_test_processed = preprocessor.transform(x_test)

# Initialize and train the Linear Regression model with processed data
model = LinearRegression()
model.fit(x_train_processed, y_train)
```

▾ LinearRegression i ?

```
LinearRegression()
```

```
y_pred=model.predict(x_test_processed)
mse=mean_squared_error(y_test,y_pred)
print(mse)
print(r2_score)
```

```
33596915.85136146
<function r2_score at 0x7a9b14f89080>
```

```
new_data_for_prediction = df[['age', 'sex', 'bmi', 'chi
processed_new_data = preprocessor.transform(new_data_fo
predicted_charges = model.predict(processed_new_data)
print('predicted_charges:',predicted_charges)
```

```
predicted_charges: [25197.53106142  3826.78192926  6987.53528962 ...  4458.33989738
  1352.45925822 36824.03102006]
```